

1 Motivation

Music is **subjective** and **one-to-many**: a sunset video may inspire a melancholic piano piece or an uplifting orchestral swell. Three limitations persist:

- ① **Paired-data bottleneck.** End-to-end training needs scarce vision/text-music pairs, capping data scale and acoustic quality.
- ② **No explicit one-to-many modeling.** Ambiguity is left to stochastic noise — no interpretable mechanism for semantic diversity.
- ③ **Generation $\perp$ retrieval.** Complementary tasks are built as separate models, wasting shared cross-modal understanding.

Key idea — a music semantic anchor. Introduce a continuous music-semantic space (a frozen **MuQ** encoder) that **decouples multimodal understanding from acoustic synthesis**. Understanding = mapping *any* modality *into* the anchor (one-to-many). Synthesis = mapping *from* the anchor to waveform. At inference the *same* predicted anchor is both the **condition** for synthesis and the **query** for retrieval.

text / image / video $\rightarrow$ Qwen3-VL-Emb $\rightarrow$ Music Head (27M) $\rightarrow$ anchor $\hat{z}_m \Rightarrow$
Flow-DiT $\rightarrow$ wave / cos-sim $\rightarrow$ Top-k

- First unified framework for both **generation and retrieval** of music from text, image and video.
- Breaks the data bottleneck** — anchor decoupling lets Stage I exploit **~630K unpaired** music clips, independent of paired data.
- Retrieval for free** — no contrastive training, no task-specific head; both tasks share one forward pass up to $\hat{z}_m$.
- SoTA or competitive** across text-to-music, vision-to-music generation and cross-modal retrieval benchmarks.

3 Two-Stage Decoupled Training

STAGE I Acoustic modeling — music-only data

A Flow-Matching DiT learns to reconstruct high-fidelity audio latents from anchor embeddings, using **large unpaired corpora** (FMA 100K + MusicSet 130K + AudioSet-Music 400K). Acoustic quality is **freed from paired-data scarcity**.

$$\mathcal{L}_I = \mathbb{E}_{m,t,z_m} \| h_\psi(x_t, z_m) - (1 - \alpha_t) \varepsilon_t \|^2$$

STAGE II Cross-modal alignment — paired data

A lightweight **diffusion** Music Head maps query embeddings into the anchor space. Because it **samples** rather than regresses, it captures genuine **one-to-many** ambiguity instead of collapsing to the conditional mean.

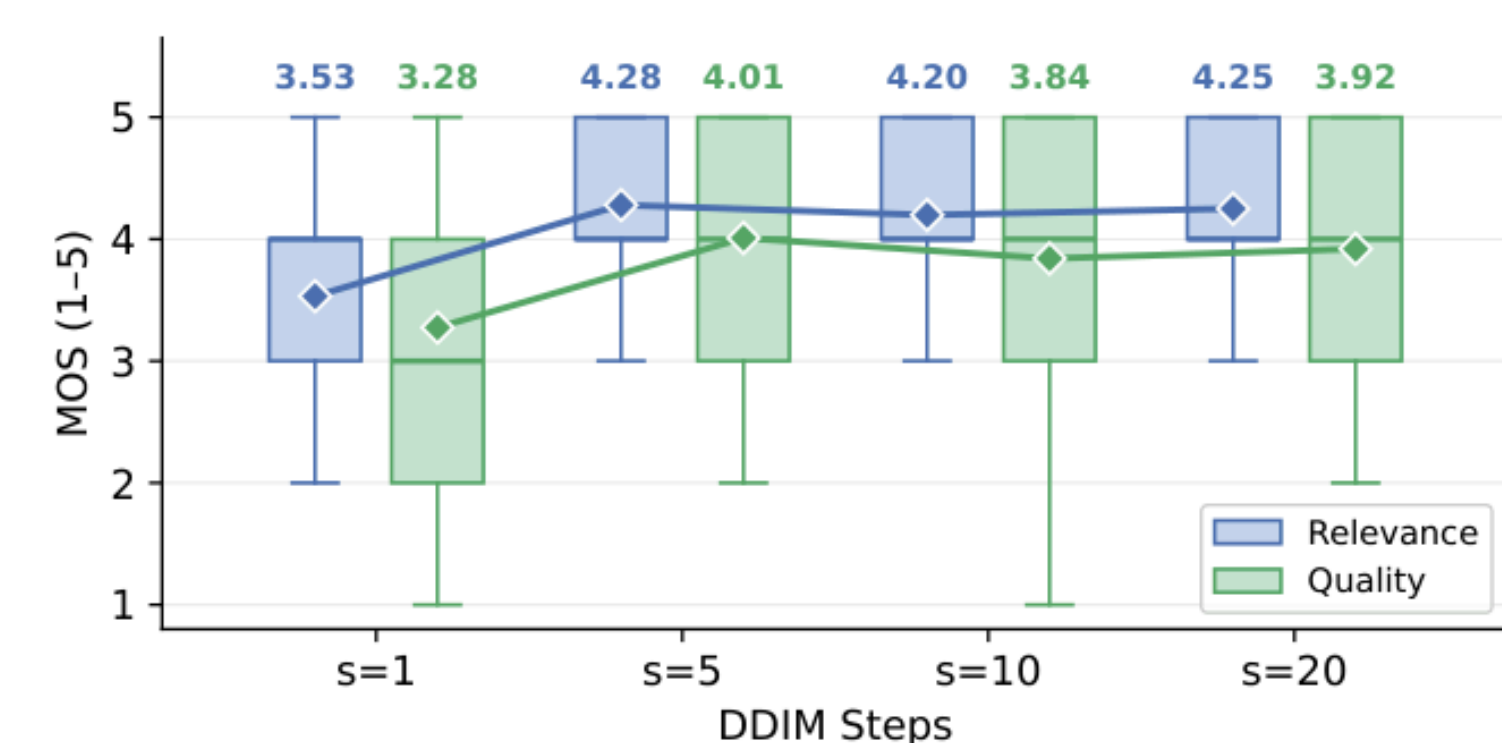
$$\mathcal{L}_{II} = \mathbb{E}_{(q,m),\varepsilon,t} \| \varepsilon - g_\psi(\sqrt{\alpha_t} z_m + \sqrt{1-\alpha_t} \varepsilon, t, e_q) \|^2$$

Why diffusion? the fidelity-diversity trade-off

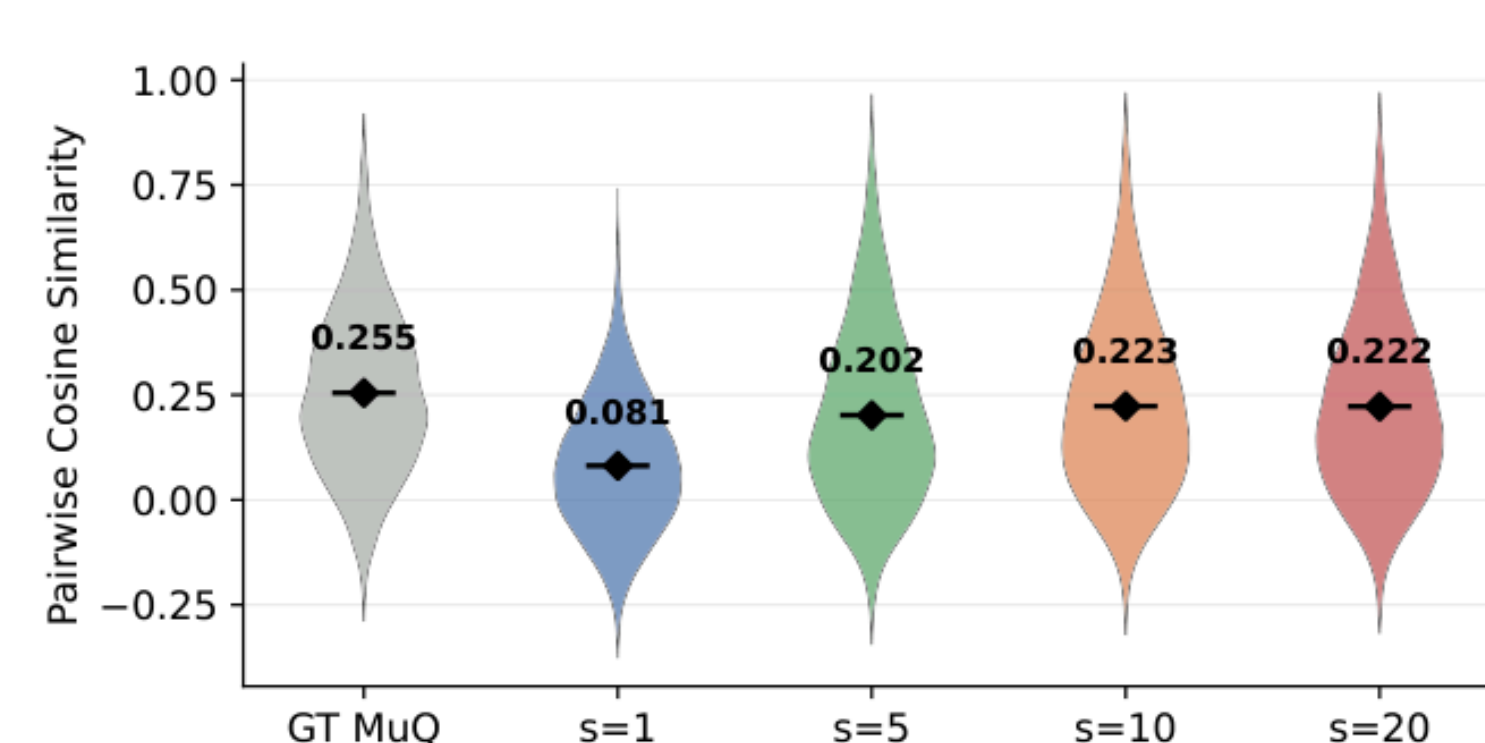
Projector	Loss	FAD _{merit} ↓		R@1 / R@10 (%) ↑	
		MelB.	ARIA	MelB.	ARIA
Linear	Contrastive	12.94	20.04	1.70 / 9.02	0.70 / 5.50
Linear	MSE	4.37	<u>7.01</u>	0.82 / 5.56	0.50 / 4.10
BridgeVAE	MSE+KL	4.06	9.35	0.84 / 6.28	0.80 / 3.50
CrossVAE	+Cos	3.48	7.22	0.98 / 5.88	0.80 / 4.30
Music Head	Diffusion	1.96	4.32	1.52 / 9.30	0.80 / 4.80

Contrastive linear retrieves well but has catastrophic FAD (6.6 $\times$); MSE/VAE variants collapse to the conditional mean. **Only iterative denoising resolves the trade-off** — predictions individually accurate yet collectively diverse.

Sampling steps buy diversity, not just quality



(a) Human MOS ratings (relevance & quality)

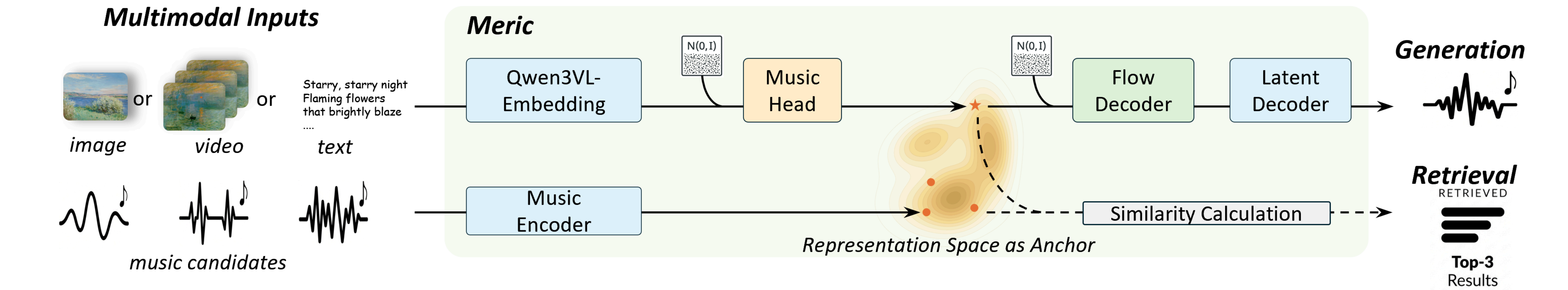


(b) Intra-group pairwise cosine similarity

Fig. 3. MOS: s=1 is clearly worse (p<0.001) but s $\geq$ 5 are indistinguishable (p>0.2). Pairwise cosine similarity explains why — few steps **collapse** the anchor distribution, while s=20 matches ground-truth MuQ diversity. **s=5 suffices at 4 $\times$ lower latency.**

2 Meric Framework — one anchor, two tasks

Inference · generation and retrieval share one forward pass



Two-stage decoupled training

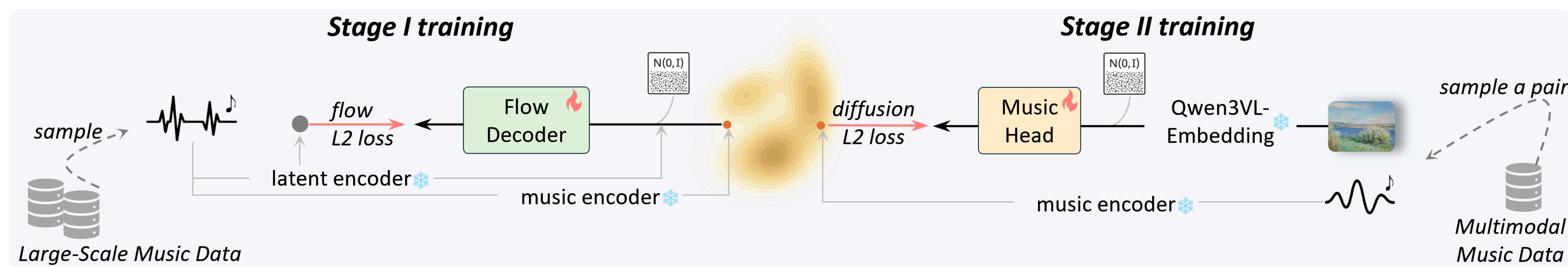


Fig. 1. A frozen Qwen3-VL-Embedding encodes any modality; the Music Head maps it to a music semantic embedding that is simultaneously decoded to audio (Flow-DiT + latent decoder) and matched against a pre-indexed music library.

4 Generation Results

Vision $\rightarrow$ Music (FAD_{merit} ↓ Sem ↑, LLM-as-judge)

Method	MelBench		Mulimage		EmoMV		ARIA	
	FAD ↓	Sem ↑	FAD ↓	Sem ↑	FAD ↓	Sem ↑	FAD ↓	Sem ↑
Art2Mus	31.60	39.1	30.75	45.3	33.86	44.3	15.05	45.3
GVMGen	<u>3.55</u>	52.9	3.78	49.6	3.96	63.6	4.18	60.0
AudioX	16.48	53.8	13.90	51.8	4.75	79.0	20.15	38.5
Qwen3VL + MusicGen	4.42	80.6	<u>2.99</u>	85.6	7.08	80.3	5.09	76.6
Qwen3VL + AudioLDM2-L	6.57	<u>77.6</u>	4.59	87.2	8.61	78.7	9.95	<u>77.6</u>
MeriC (Ours)	1.55	74.7	1.79	87.1	3.06	81.0	3.30	79.6

Lowest FAD on all four benchmarks (also FAD_{clap}, FAD_{vgg}), while matching or beating strong caption-cascades on semantics and exceeding every direct baseline by >20 Sem points.

Text $\rightarrow$ Music — zero text-audio training

Method	MusicCaps			ARIA		
	FAD _{merit} ↓	FAD _{clap} ↓	Sem ↑	FAD _{merit} ↓	FAD _{clap} ↓	Sem ↑
MusicGen-Medium	3.80	0.421	65.1	3.00	0.272	71.1
MusicGen-Large	3.51	0.409	67.1	<u>2.55</u>	0.248	72.4
AudioLDM2-Music	4.60	0.260	68.0	4.63	0.251	68.6
AudioLDM2-Large	4.17	<u>0.178</u>	76.7	3.93	<u>0.211</u>	77.5
Stable-Audio-Open	3.85	0.268	68.3	3.50	0.324	57.8
MeriC (Ours)	1.87	0.110	74.2	2.14	0.155	74.5

Qwen3-VL-Emb places text and images in one space, so **text-to-music emerges for free** — yet Meric still beats every dedicated T2M model on FAD.

Vision encoder matters

Backbone	Dim	FAD _{merit} ↓	
		MelB.	ARIA
CLIP-ViT-H-14	1024	2.54	4.73
Qwen3VLE	2048	1.95	3.98

Richer vision-language representations help cross-modal translation even when frozen.

New dataset — ARIA (Art-Referenced Instrumental Audio): **5K images**, 50% Unsplash photographs / 50% WikiArt artworks, each captioned 3 $\times$ by Gemini 3 Flash and paired with 3 instrumental tracks (~9 $\times$ 10 s segments per image; 4K/1K train-test split). It is the benchmark where generic contrastive embeddings fall down and the **anchor space wins in every direction**.

5 Retrieval — a Zero-Cost Byproduct

The Music Head already maps queries into the anchor space, so ranking is plain **cosine similarity** against a pre-indexed library — **no contrastive training at all**.

Vision $\leftrightarrow$ Music

Method	MelBench (1-to-1)		ARIA (1-to-many)	
	I $\rightarrow$ M	M $\rightarrow$ I	I $\rightarrow$ M	M $\rightarrow$ I
ImageBind	4.38 / 18.0	3.66 / 17.0	0.30 / 2.30	0.13 / 1.77
MeriC (Ours)	1.34 / 8.62	1.12 / 9.54	0.40 / 2.80	0.39 / 2.40

Method	MusicCaps		ARIA	
	T $\rightarrow$ M	M $\rightarrow$ T	T $\rightarrow$ M	M $\rightarrow$ T
CLAP (630K pairs)	4.31 / 19.5	5.54 / 23.5	0.17 / 2.30	0.63 / 3.73
MeriC (Ours)	3.27 / 18.8	2.82 / 16.4	0.83 / 5.47	0.73 / 6.40

R@1 / R@10 (%). Generic contrastive models lead on their home turf, but on the **art-referenced ARIA** benchmark Meric wins in **every direction** — the anchor space captures fine-grained artistic correspondences that generic embeddings miss (T $\rightarrow$ M R@1: **0.83** vs 0.17).

6 LLM-as-Judge (Sem)

FAD cannot tell whether music *semantically matches* its visual input. We prompt **Gemini 3 Flash** with a 4-step protocol — scene description $\rightarrow$ audio attributes $\rightarrow$ alignment analysis $\rightarrow$ calibrated score $\in$ [0,100] — and validate it on a **300-task, 6-way forced-choice** benchmark against 8 human annotators.

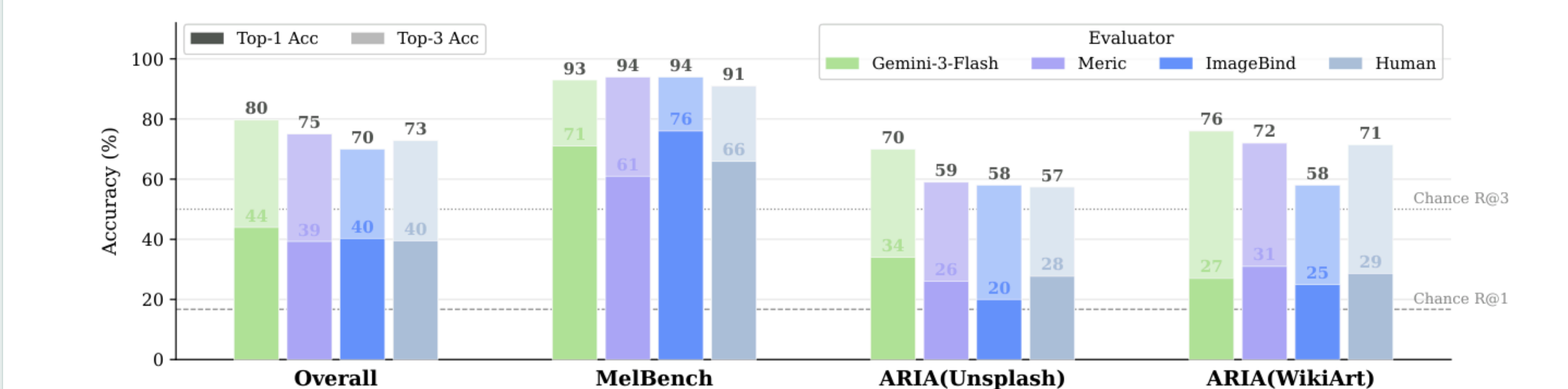


Fig. 2. ImageBind leads on MelBench, but Gemini wins the harder artistic domains and takes the **best overall accuracy (44.0%, MRR 0.644)** — exceeding human annotators — while tracking the same difficulty pattern as humans. This validates Sem as our semantic metric.

Takeaway. Anchoring on a music semantic space turns three separate problems into one design choice: unpaired corpora lift acoustic quality, a diffusion head makes ambiguity *explicit and samplable*, and retrieval falls out of the same forward pass.

